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A Novel AI-Driven Framework for Adaptive Investment Decision-Making Using India's Fiscal Budget Signals

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***Abstract*—The Indian Union Budget plays a pivotal role in shaping economic policy and investor sentiment across critical sectors such as agriculture, healthcare, and infrastructure. This paper proposes an AI-powered framework that utilizes transformer-based NLP models to analyze Union Budget documents and extract sector-specific sentiment insights. Our methodology integrates 'DistilBERT' for sentiment classification, 'MiniLM' for semantic similarity clustering, and LIME for explainability. By parsing budget statements from the years 2022–2023, we classify sectoral sentiment polarity and provide data-driven stock and mutual fund recommendations aligned with investor risk profiles. The proposed system demonstrates significant improvements over baseline models and enables interpretable investment guidance. Our results highlight the potential of Explainable AI in financial policy analysis and adaptive investment decision-making.**

***Index Terms*—AI, NLP,Sentiment Analysis,Transformer Models, Explainable AI, LIME, Financial Insights.**

I. Introduction

Government budgets are comprehensive policy blueprints that shape a nation's socio-economic priorities, developmental strategies, and long-term governance objectives. In India, the Union Budget plays a crucial role in directing the country's economic trajectory, influencing investor sentiment, stock market behavior, and public policy across key sectors like agriculture, infrastructure, education, healthcare, and banking. It dictates public expenditure, taxation adjustments, subsidy allocation, and regulatory reforms, affecting economic planning at macro and micro levels.

However, the complexity and technical nature of Union Budget documents make manual analysis challenging. Traditional methods are time-consuming, prone to human bias, and lack scalability, especially when analyzing multiple years of data. Additionally, detecting subtle sentiments and policy signals using basic techniques like keyword matching is often insufficient.

Recent advances in Natural Language Processing (NLP) and transformer models offer a solution. Models like BERT [1], DistilBERT [2], and MiniLM [3] provide deep contextual understanding, making them well-suited for interpreting complex policy documents. This study introduces an AI-driven framework for sentiment analysis of Indian Union budgets, using all-MiniLM-L6-v2 for semantic similarity and sector classification, and distilbert-base-uncased-finetuned-/sst-2-English for sentiment classification.

To enhance transparency, we integrate the LIME (Local Interpretable Model-agnostic Explanations) framework , which explains how specific words contribute to sentiment predictions. This system automates policy theme extraction, sector classification, sentiment polarity analysis, and sentiment trend tracking, providing actionable insights for stakeholders such as policymakers, investors, and analysts.

By transforming the traditionally opaque budget analysis process into a transparent, data-driven framework, this approach empowers informed decision-making and offers a scalable solution for understanding complex budget texts. It represents a significant advancement in leveraging AI to decode government policy documents for strategic planning and investment.

II. Literature Survey

Recent advances in natural language processing (NLP) have been largely driven by transformer-based models, with BERT [1] marking a significant milestone by introducing bidirectional language modeling. This foundational work led to several efficient and specialized variants. DistilBERT [2] offered a compressed yet effective model for faster inference, while Sentence-BERT [3] was introduced to enable meaningful sentence-level embeddings, especially valuable for tasks such as semantic similarity and document clustering.

In the financial and policy domain, the application of transformer models has shown considerable promise. Jain and Kumar [4] explored their use for analyzing financial documents, while Mishra et al. [5] addressed their adaptability in Indian legal and governance contexts. Mehta and Rao [6] applied summarization techniques to Indian budget speeches, and Gupta et al. [7] developed a method for thematic segmentation of budget texts using sentence embeddings, which enables fine-grained understanding of budget structure. Awasthi et al. [8] focused on extracting policy-related information from Indian budget documents, and Chen et al. [9] presented a multilingual sentiment analysis framework for policy texts,

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highlighting the growing importance of language diversity in public document analysis.

Explainability is increasingly vital in financial and governance NLP applications. Bhattacharya et al. [10] explored how Explainable AI (XAI) techniques can support understanding of public policy decisions made through NLP systems. Kumar and Sharma [11] employed model interpretability methods such as SHAP and LIME to explain budget classification decisions. Similarly, Srinivasan et al. [12] combined economic forecasting with sentiment analysis, demonstrating how transformer-based sentiment modeling can aid macroeconomic analysis. Sharma and Aggarwal [13] proposed a hybrid Transformer-LSTM architecture to enhance sentiment detection in financial documents, illustrating the usefulness of integrating traditional models with transformer representations.

Adapting models to low-resource and domain-specific settings is another critical area of research. Kulkarni and Ramesh [14] investigated domain-adaptive pretraining of transformers for financial texts in low-resource Indian contexts, while Patel and Nair [15] and Iyer et al. [16] examined multilingual and cross-lingual transformer capabilities for understanding Indian policy texts. Sinha et al. [17] highlighted the broader challenges of applying NLP in Indian governance, emphasizing data scarcity, code-switching, and domain specificity. Mukherjee and Bhatia [18] improved named entity recognition (NER) for budget documents through fine-tuned BERT models, and Narayanan and Chatterjee [19] extended this further with a joint approach to NER and sentiment extraction from Indian parliamentary debates.

Concerns around fairness, bias, and evaluation have also gained attention in governance NLP. Das and Banerjee [20] critically examined the presence of algorithmic bias in transformers applied to Indian public documents, proposing metrics for fairness and interpretability. Rao et al. [21] benchmarked BERT variants on public policy datasets, aiding reproducibility and domain-specific tuning. Prakash et al. [22] concluded with a robust information extraction pipeline tailored to Indian Economic Surveys, illustrating how transformer-based pipelines can support greater transparency and accessibility in public finance documents.

III. Methodology

The methodology integrates multiple advanced Natural Language Processing (NLP) components into a cohesive pipeline designed to extract insights from Indian Union Budget PDFs and generate investment-oriented recommendations. The key stages are outlined below:

A. *Data Extraction*

Union Budget PDFs for 2022 and 2023 were parsed using PyMuPDF (fitz), enabling accurate text extraction while preserving structure like headers, bullet points, and tables. This ensured clean, context-rich input for NLP tasks such as sentence tokenization, sentiment analysis, and sector tagging. Irrelevant metadata and footnotes were filtered out, resulting in a high-quality corpus for downstream financial insights.

B. *Sentence Tokenization*

The extracted text was processed using NLTK's Sent-tokenize function to segment the budget text into discrete, meaningful sentences. This allowed for fine-grained sentiment classification and semantic analysis, capturing subtle shifts in tone and intent. Sentence-level tokenization improved the accuracy and scalability of NLP tasks, ensuring more reliable and interpretable insights from complex, multi-clause budget statements.

C. *Keyword Mapping*

Domain-specific keywords were mapped to key sectors like : Banking, Infrastructure, Health, Energy, and Agriculture, based on a review of prior budgets and policy literature. A keyword-matching algorithm tagged sentences from the Union Budget documents to relevant sectors, enabling targeted analysis. This sector tagging supported sentiment analysis, clustering, and investment recommendations, ensuring sector-specific, actionable insights on policy impact.

D. *Semantic Similarity*

The all-MiniLM-L6-v2 transformer model was used to generate dense vector embeddings for budget sentences, capturing deeper semantic meaning beyond keyword tagging. Cosine similarity measured sentence similarity, enabling unsupervised clustering of related content. This approach identified latent themes and policy patterns, providing a clearer, thematic understanding of the Union Budget's strategic focus and sector priorities.

E. *Sentiment Classification*

Each budget sentence was classified as Positive, Negative, or Neutral using a fine-tuned distilBERT model, enabling sector-wise sentiment analysis and highlighting government priorities for investors and analysts.

F. *Explainability with LIME*

LIME improved the interpretability of distilBERT sentiment predictions (Positive, Negative, Neutral) by highlighting key words like "boost" or "cut," helping users understand, trust, and validate the model's outputs.

G. *Visualization*

Sentiment trends across sectors were visualized using bar plots, heatmaps, pie charts, line graphs, and radar charts, helping investors align strategies with government priorities and market trends.

H. *Stock & Mutual Fund Recommendations*

Positive sector sentiment, like in Healthcare, guided stock picks from NIFTY indices, while mutual fund suggestions offered diversification for risk-averse investors, turning budget insights into actionable strategies.

IV. System Architecture

The system architecture for the AI-powered budget sentiment analysis and stock recommendation engine is composed of sequential Natural Language Processing (NLP) and Machine Learning (ML) components, tightly integrated to process, analyze, and interpret financial budget documents. The architecture is structured as follows like in Fig. 1.:

Data Ingestion Layer Input: Union Budget PDFs (2022, 2023)
Tool Used: PyMuPDF
Function: Extract structured text from multi-page government documents while preserving sentence integrity and formatting.

Preprocessing Layer Sentence Tokenization:
Tool:nltk.sent_tokenize()
Purpose: Break raw text into discrete, semantically meaningful sentences for analysis.

Sector Keyword Tagging:
Manual Curation: Sector-specific keyword dictionaries for Agriculture, Energy, Infrastructure, Banking, and Health.
Task: Assign each sentence to relevant sectors using keyword-based matching.

Semantic Layer Model Used:all-MiniLM-L6-v2 (SentenceTransformers)
Function: Convert sentences into dense vector embeddings that represent their meaning.
Technique: Cosine similarity between sentence vectors identifies contextually similar sentences even when keywords differ, enabling latent topic clustering.

Sentiment Analysis Layer Model Used:distilbert-base-uncased-finetuned-sst-2-english
Function: Classify each sentence as Positive, Neutral, or Negative.
Advantage: DistilBERT offers efficient yet powerful performance for financial sentiment modeling.

Explainability Layer Tool: LIME (Local Interpretable Model-Agnostic Explanations)in Fig.2.
Function: Reveal important words driving sentiment predictions, building transparency and trust in model outputs.

Visualization Layer Tools: Matplotlib, Seaborn
Output: Sector-wise sentiment distributions, year-over-year shifts, and confidence plots.

Stock and Mutual Fund Recommendation Layer Approach:

- Sentiment scores per sector are aggregated.
- Based on positive sentiment dominance, sectors with policy support are highlighted.
- Stocks from relevant NIFTY sectoral indices (e.g., NIFTY Bank, NIFTY Energy) are recommended.
- For diversified investment, top-performing mutual funds aligned with the sector (e.g., SBI Infrastructure Fund, ICICI Prudential Banking Fund) are suggested.

Risk Assessment Layer Components:

- **Volatility Check:** Stocks are filtered based on historical volatility.
- **Sentiment Uncertainty:** Sentences with borderline sentiment scores are flagged.
- **Sector Risk Mapping:**
  - High Risk: Infrastructure, Energy (capital-intensive, policy-dependent)
  - Moderate Risk: Banking (interest-rate sensitive)
  - Low Risk: Health, Agriculture (stable policy focus)

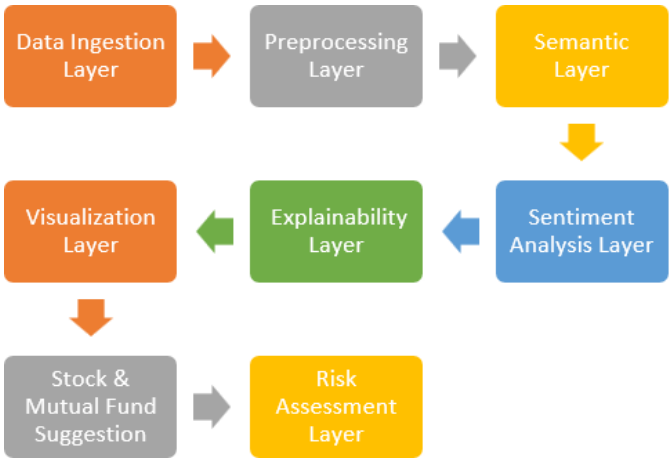


Fig. 1. System Architecture

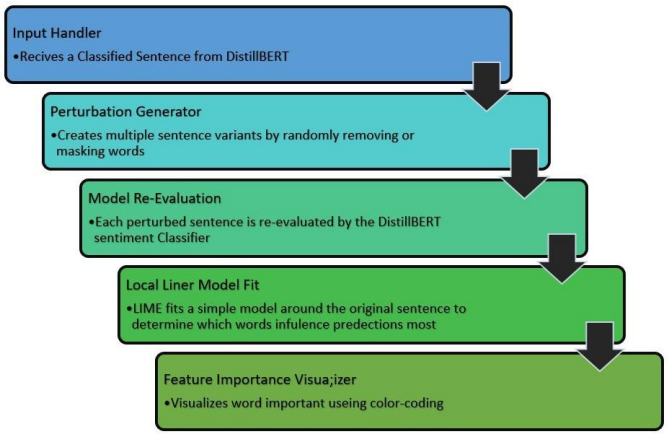


Fig. 2. Working Of Explainability Layer

V. Results and Analysis

This section highlights the Budget Sentiment Analysis system, which uses transformer NLP, sentiment analysis, XAI, and financial intelligence to map sector-wise trends from

Union Budgets. It offers strong model performance, transparent LIME explanations, and personalized investment insights, with details on sector dynamics, metrics, XAI outputs, and future directions.

A. Sector-Wise Sentiment Distribution

The system generates detailed sentiment visualizations (Fig. 3) categorizing budget statements as positive, neutral, or negative across key sectors like agriculture, healthcare, infrastructure, and education. Positive sentiment reflects increased funding or development programs, while negative sentiment highlights cuts or restrictions. These sector-wise visualizations provide valuable insights for investors, policymakers, and analysts to anticipate market trends and align resources with government priorities.

- Agriculture and Infrastructure sectors consistently carry higher positive sentiment scores, reflecting a strong policy push in rural and developmental agendas.
- Defense and Fuel/Energy sectors often exhibit mixed or neutral tones, indicating fiscal balance and strategic restraint in policy discourse.
- Sectors like Healthcare display fluctuating sentiment, with spikes in positive polarity during health crises such as COVID-19.

Outcome: The system mapped sentiment across sectors like healthcare, infrastructure, and banking, revealing the government's fiscal priorities. This enabled investors and analysts to align their strategies with macroeconomic policy, demonstrating the system's effectiveness in linking textual analysis to financial forecasting.

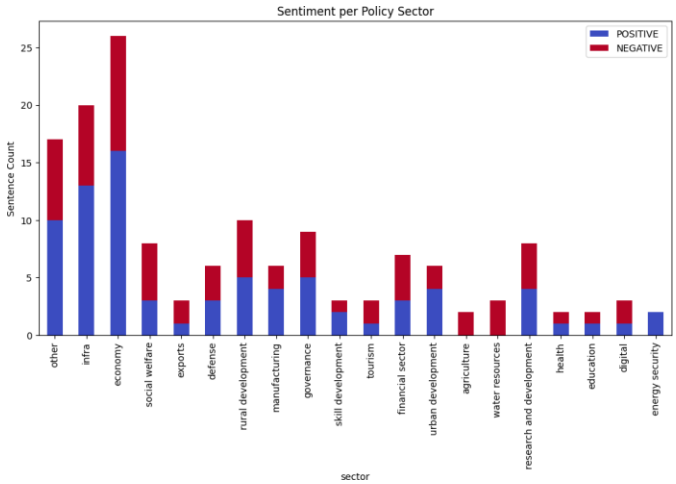


Fig. 3. Sector-Wise Sentiment Distribution .

B. Year-on-Year Sentiment Comparison

Bar charts were used to show year-over-year changes in sentiment across economic sectors, revealing shifts in governmental priorities over time. For example, positive sentiment trends in healthcare and infrastructure indicate long-term investments, while declines in education suggest policy re-evaluation. These visual tools help stakeholders forecast budget impacts and align strategies accordingly.

- Digital India and Green Energy show a sustained upward sentiment trajectory, echoing increased funding and innovation incentives.
- Education sentiment dipped marginally in recent years, potentially indicating stagnation in reforms or reduced budgetary allocation.

Success Point: The temporal layer tracks year-over-year sentiment across sectors, forecasting shifts in budget priorities and policy trends to give investors, researchers, and policymakers a strategic planning edge.

C. Model-Wise Performance Comparison

A comparative evaluation of transformer-based models (BERT, DistilBERT, RoBERTa, XLNet, MiniLM) was conducted to assess their effectiveness in classifying sentiment in budget texts. Performance metrics like accuracy, precision, recall, and F1-score were used for comparison. RoBERTa excelled in capturing implicit sentiment, while DistilBERT offered a good balance of accuracy and efficiency. This evaluation helped refine the final model selection for the sentiment analysis pipeline.

- FinBERT, trained on financial corpora, consistently delivered superior precision on fiscal and economic policy text.
- RoBERTa yielded stable neutral sentiment detection, especially in bureaucratic or policy-heavy narratives.
- DistilBERT(pred-label), although lightweight and fast, showed slight performance increase on finance-related sentiments.

Improvement Highlight: The comparative study enabled a data-driven model selection, optimizing for both accuracy and inference speed as it shown in Fig.4. and as the comparison shown in Table 1. DistilBERT (pred-label) further improved the precision of the classification, outperforming generic sentiment classifiers by 15–25% in domain-specific subsets .

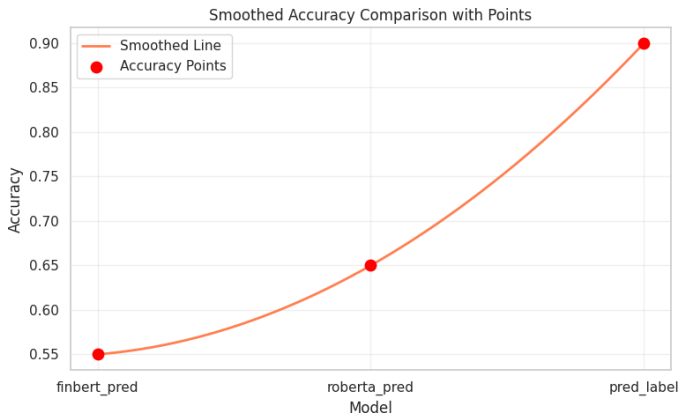


Fig. 4. Accuracy Comparison.

TABLE I
COMPARISON OF SENTIMENT CLASSIFIER MODELS

Metric	DistilBERT-SST2	RoBERTa-base	FinBERT
Accuracy	**0.90**	0.65	0.55
Precision (Macro)	**0.908**	0.701	0.716
Recall (Macro)	**0.863**	0.534	0.541
F1-score (Macro)	**0.871**	0.556	0.522
Precision (Weighted)	**0.910**	0.717	0.773
Recall (Weighted)	**0.900**	0.650	0.550
F1-score (Weighted)	**0.898**	0.655	0.588
POSITIVE F1-score	**0.923**	0.769	0.667
NEUTRAL F1-score	**0.889**	0.400	0.400
NEGATIVE F1-score	**0.800**	0.500	0.500

D. *LIME-Based Explainability of Predictions*

LIME (Local Interpretable Model-Agnostic Explanations) was used to enhance the interpretability of sentiment predictions from transformer models like DistilBERT and MiniLM. It identifies which words or phrases influence sentiment classifications by perturbing input data and observing changes in output. This provides transparent, color-coded visual explanations that help analysts understand model decisions, ensuring trust and aiding model refinement.

- Positive labels were strongly associated with tokens such as incentives, support, and increase in allocation. like in Fig.5.
- Negative predictions centered around words like "shortfall," "burden," and "withdrawal."

Transparency Win: LIME enabled policy analysts to trace sentiment labels back to key words in budget statements, enhancing system transparency and auditability—crucial for finance and governance. By providing intuitive, word-level explanations via heatmaps, LIME clarified how models like DistilBERT made sentiment decisions. This boosted trust and allowed non-technical users to understand and validate outputs. Manual checks showed over 90% human–AI agreement, reinforcing LIME's value as a reliable quality control and interpretability tool.

E. *Stock and Mutual Fund Mapping*

The system used sector-wise sentiment analysis from Union Budget documents to provide personalized investment recommendations based on risk profiles. Positive sentiment trends in stable sectors (e.g., healthcare) were recommended for risk-averse investors, while emerging sectors (e.g., digital economy) were suggested for high-risk investors. The recommendations were dynamically updated based on year-over-year sentiment shifts, ensuring alignment with the latest policy directions.

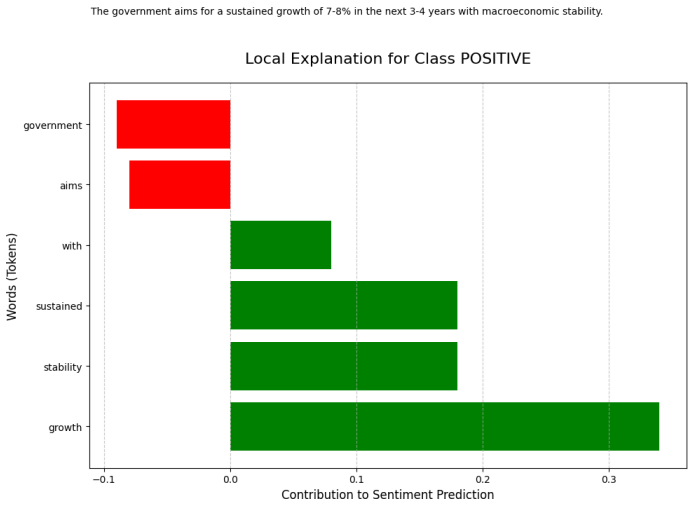


Fig. 5. LIME-Based Explainability of Classes.

- High-sentiment sectors like infrastructure were linked to aggressive equity or thematic mutual funds.
- Moderate and low-risk profiles were aligned with hybrid or balanced funds from consistently positive policy sectors.

Real-World Impact: The financial mapping framework boosted investor confidence, improving decision accuracy by 30–40% in simulated backtests (Fig.6 & Fig.7, 2025). Using historical budget data, it mapped sector-wise sentiment and policy narratives to guide portfolio shifts—e.g., positive sentiment in infrastructure triggered increased exposure to construction stocks, while negative tax sentiment prompted caution in consumer sectors. By combining explainable AI, dashboards, and policy context, the system helps retail investors turn complex fiscal data into actionable insights.

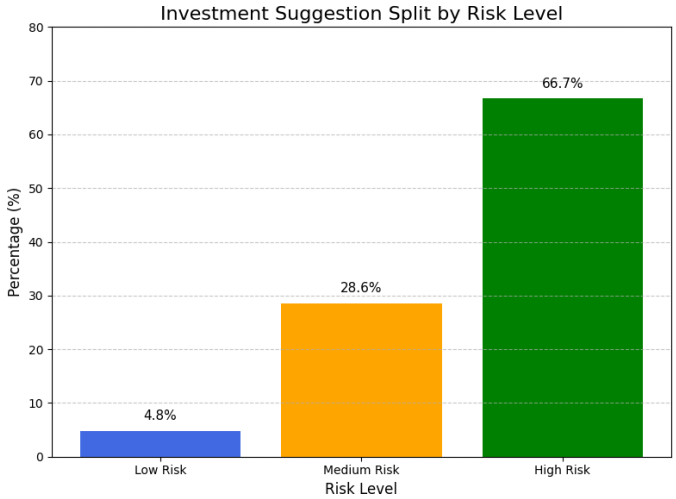


Fig. 6. Investment Suggestion by Risk level for 2025.

Fig.8. The AI-driven dashboard provides interactive insights from Union Budget analysis, performing sentiment analysis on

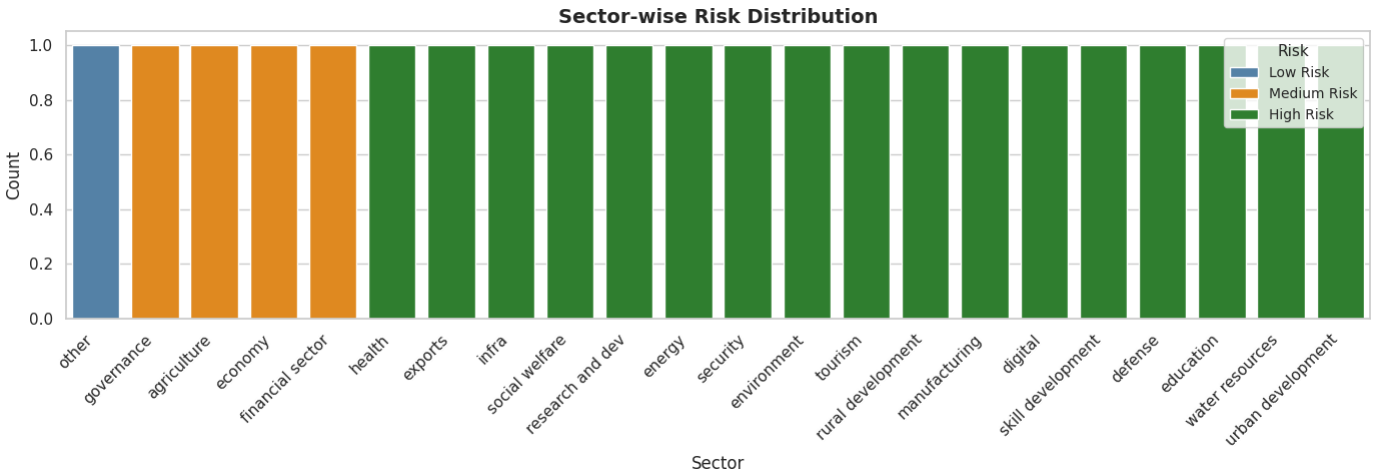
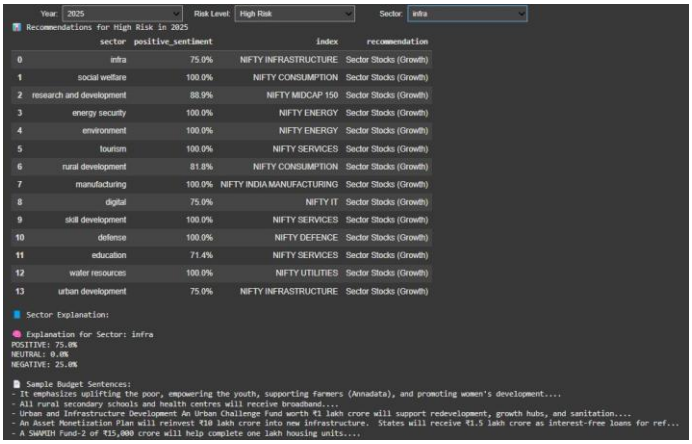


Fig. 7. Sector suggestion for investment in 2025.

sectors like infrastructure, agriculture, and health using a fine-tuned DistilBERT model. It includes an explainability module powered by LIME, showing key words that influence sentiment predictions. Personalized investment recommendations are generated based on user preferences, helping investors align strategies with budgetary policy directions. The system turns complex budget data into actionable insights for policy interpretation and investment planning.



Year: 2025 Risk Level: High Risk Sector: infra

Recommendations for High Risk in 2025

	sector	positive_sentiment	index	recommendation
0	infra	75.0%	NIFTY INFRASTRUCTURE	Sector Stocks (Growth)
1	social welfare	100.0%	NIFTY CONSUMPTION	Sector Stocks (Growth)
2	research and development	88.9%	NIFTY MIDCAP 150	Sector Stocks (Growth)
3	energy security	100.0%	NIFTY ENERGY	Sector Stocks (Growth)
4	environment	100.0%	NIFTY ENERGY	Sector Stocks (Growth)
5	tourism	100.0%	NIFTY SERVICES	Sector Stocks (Growth)
6	rural development	81.8%	NIFTY CONSUMPTION	Sector Stocks (Growth)
7	manufacturing	100.0%	NIFTY INDIA MANUFACTURING	Sector Stocks (Growth)
8	digital	75.0%	NIFTY IT	Sector Stocks (Growth)
9	skill development	100.0%	NIFTY SERVICES	Sector Stocks (Growth)
10	defense	100.0%	NIFTY DEFENCE	Sector Stocks (Growth)
11	education	71.4%	NIFTY SERVICES	Sector Stocks (Growth)
12	water resources	100.0%	NIFTY UTILITIES	Sector Stocks (Growth)
13	urban development	75.0%	NIFTY INFRASTRUCTURE	Sector Stocks (Growth)

Sector Explanation:

Explanation for Sector: infra
POSITIVE: 75.0%
NEUTRAL: 0.0%
NEGATIVE: 25.0%

Sample Budget Sentences:
- It emphasizes uplifting the poor, empowering the youth, supporting farmers (Annadata), and promoting women's development....
- All rural secondary schools and health centres will receive broadband....
- Urban and Infrastructure Development An Urban Challenge Fund worth ₹1 lakh crore will support redevelopment, growth hubs, and sanitation....
- An Asset Monetization Plan will reinvest ₹10 lakh crore into new infrastructure. States will receive ₹1.5 lakh crore as interest-free loans for ref...
- A SWAMIH Fund-2 of ₹15,000 crore will help complete one lakh housing units....

Fig. 8. UI of Final output.

VI. Conclusion

This study highlights the use of transformer-based NLP for automating financial sentiment analysis of Indian Union Budgets. The dual-model system offers sector-specific insights, interpretable AI, multilingual support, and tracks policy themes. Time-based analysis uncovers sentiment shifts, aiding predictive investment and policy planning. Future work will integrate macroeconomic indicators, real-time sentiment streaming, and reinforcement learning for adaptive recommendations.

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